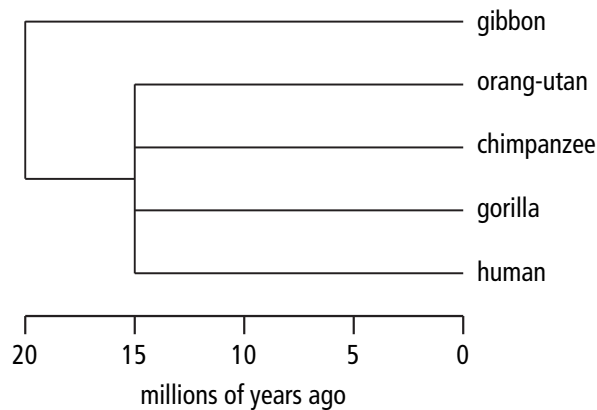
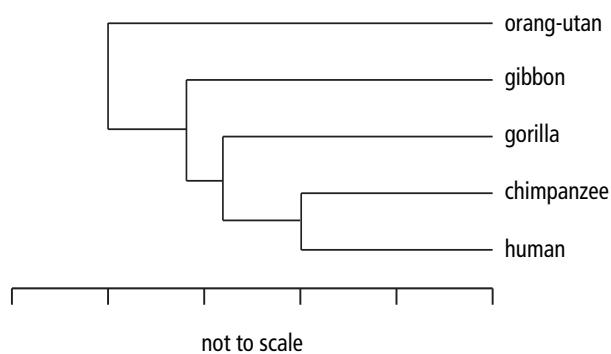


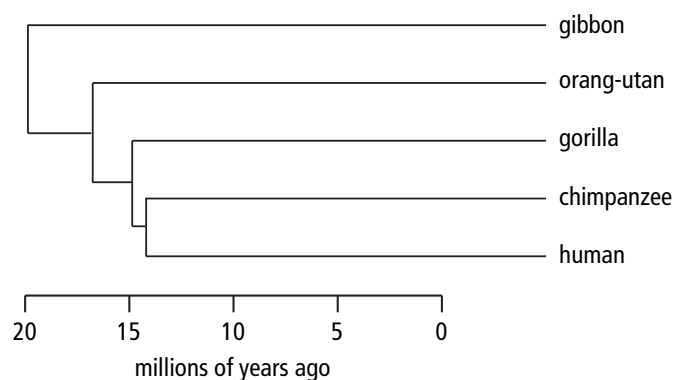
Figure 1 shows three evolutionary trees for Hominoidea. Tree A is a traditional evolutionary tree that was developed before molecular evidence was available; tree B is the evolutionary tree, based on comparisons of the 42-nucleotide cytochrome *b* segment, you built in the learning object; and tree C is based on the full cytochrome *b* sequence of 1140 nucleotides.



tree A – traditional evolutionary tree



tree B – based on 42-nucleotide segment of cytochrome *b*



tree C – based on full 1140-nucleotide cytochrome *b* sequence

Figure 1: three versions of a primate evolutionary tree

1. Outline differences between the three trees in Figure 1.

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2. Comment on **why** you think there are differences between the three trees.

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3. The most current understanding of evolutionary relationship between Hominoidea is that represented in tree C. Why do you think this evolutionary relationship is now accepted?

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4. The three evolutionary trees shown in Figure 4 look different from each other. Do the trees show the same relationships between the organisms? Identify tips, nodes and branches in tree 2.

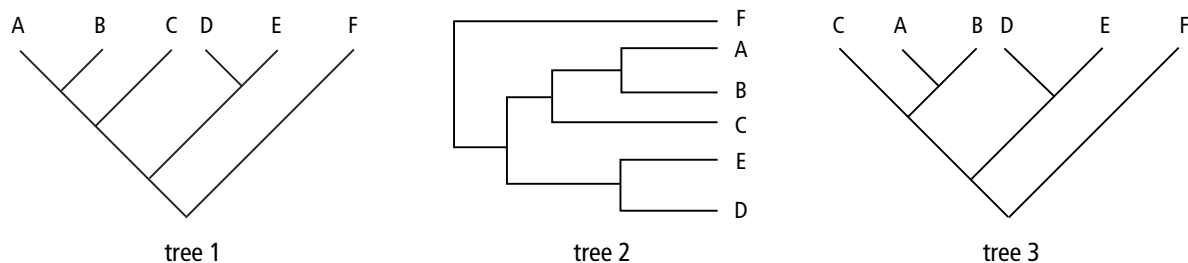


Figure 4: three evolutionary trees

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5. Figure 5 shows a full primate evolutionary tree. Mark the following on the tree:
- Label the most recent common ancestor (MRCA) of the Cercopithecinae (macaque) clade and Hominidae (great ape) clade with 'A'.
 - Label the MRCA of Hominidae and Hylobatidae (gibbon) with 'B'.
 - Label the MRCA of Cebidae (squirrel monkey) and Hylobatidae with 'C'.
 - Which are most closely related to Hominidae: Colobinae or Cebidae?
 - Which are most closely related to Atelidae (howler monkey): Hylobatidae or Lemuridae (lemur)?

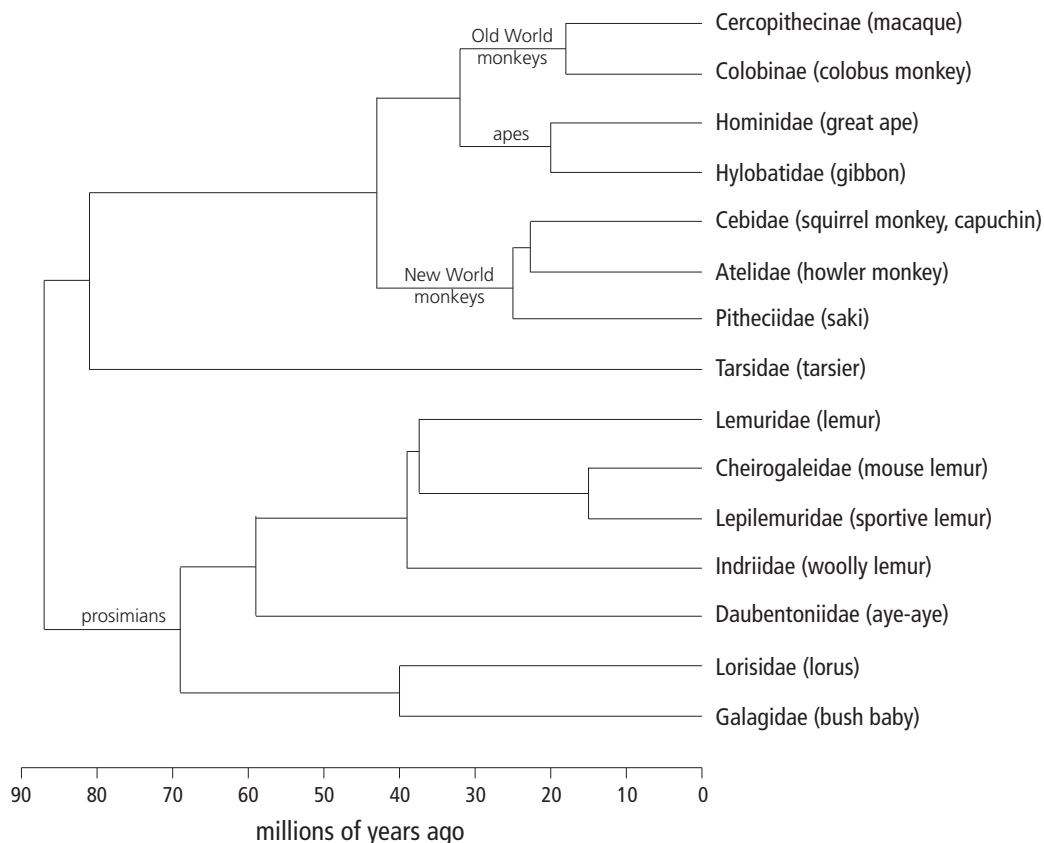
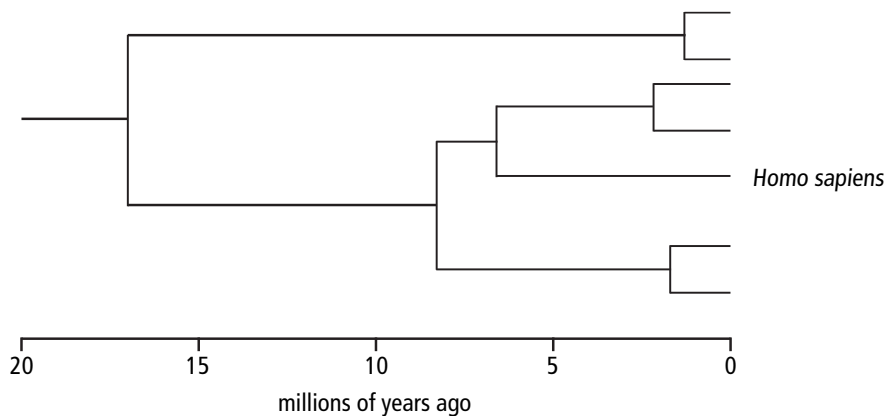
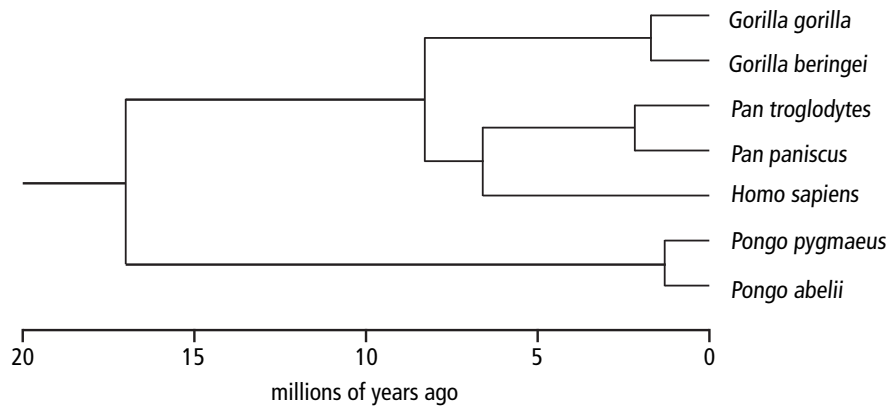


Figure 5: primate evolutionary tree

6. The order Primates forms a clade. Nested within this clade are other smaller clades that group primates according to shared characteristics. All primates share similarities, but some share particular features that others don't have. Circle the following four clades on Figure 5.

CLADE	MEMBERS	SHARED CHARACTERISTICS
Catarrhini	old world monkeys and apes	downward-facing nostrils, dry nose, large brain, sexual dimorphism
Strepsirrhini	prosimians	wet nose, protruding snout, tooth comb, upper lip attached to gum
Platyrrhini	new world monkeys	outward-facing nostrils, dry nose, opposable big toe, long tail, often prehensile
Haplorhini	old world monkeys, new world monkeys, tarsiers and apes	dry nose, larger brain, reduced snout, upper lip not attached to gum

7. Figure 6a shows an evolutionary tree for the Hominidae clade. In Figure 6b the same tree has been rearranged. Label the tree tips in Figure 6b so that evolutionary relationships between clade members remain the same.



8. Outline advantages and disadvantages of using cytochrome *b* to study evolutionary relationships, between species. Hint: Read the fact sheet, *What is molecular evidence?*

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9. Consider the basic profile of the genes listed below in Table 2. Decide which gene you think would be a potential candidate gene to study evolutionary relationships among primates. Give a reason for your decision.

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GENE	nDNA	mtDNA	BIOLOGICAL FUNCTION
EVPL (envoplaken)	yes	no	protein coding: involved in development of knuckle pads
EIF2B2 (eukaryotic translation initiation factor 2B subunit 2B)	yes	no	protein coding: involved in the initiation and control of protein synthesis
ND4 (NADH dehydrogenase 4)	no	yes	protein coding: involved in energy production (mitochondria)

Table 2: potential candidate genes for evolutionary analysis

10. A common misconception within the wider community, when discussing evolution, is that humans evolved from chimpanzees. How would you explain what is really going on?

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11. In recent decades some countries, including the European Union, Japan, Australia and New Zealand, have imposed a ban on the use of Hominidae ('great apes') in research. As well as humans, this taxonomic family includes chimpanzee, bonobo, gorilla and orang-utan.

Other countries (Spain, Austria) have taken a step further and proposed that all Hominidae be granted legal rights. This would provide them the same basic protections afforded humans, including: the right to life; freedom from exploitation and torture, and the right of consent in the participation in scientific research.

- a) How might molecular evidence for evolution contribute to future changes to animal welfare laws? What role might this evidence play in proposed legislation granting all Hominidae legal rights?

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- b) Consider ethical concerns, and comment on advantages and disadvantages of using great apes in scientific research.

Advantages:

Disadvantages:

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The primate evolutionary tree in Figure 5 is based on Perelman, P., Johnson, W.E., Roos, C. et al., 2011. A molecular phylogeny of living primates. *Plos Genetics*, 7(3), 1-17. Accessed from <http://www.plosgenetics.org>

For further information on issues raised in question 12, visit the Great Ape Project, www.greatapeproject.org